

FOR EXECUTIVES · ARCHITECTS · AI PROGRAM LEADS

THE AI-NATIVE DATA PLANE. ONE ENGINE. EVERY SIGNAL.

An architecture brief for the data platform that turns the retrieval stack — full-text, vectors, columnar logs, and agent memory — into a single Elasticsearch-compatible engine. Built in Rust. Shipped as one binary. Deployed in your cloud, your rack, or behind your air-gap.

RELEASE V1.0.0-RC.1 · ES 8.13 WIRE-COMPATIBLE · SELF-HOSTED · AGENT-READY

02 · WHY ARCHITECTURE MATTERS NOW

EVERY ENTERPRISE IS BECOMING A RETRIEVAL ENGINE.

The shift from ten-blue-link search to agentic, retrieval-augmented applications is not a UI change. It is an infrastructure change. Every product team is now standing up the same primitive: *find the right context, in the right form, fast enough for a model to use it*. Today they assemble that primitive from four vendors and a Python orchestrator. The result is a retrieval stack that is brittle, expensive, and impossible to certify.

FRAGMENTATION TAX

FOUR STORES. ONE QUERY.

A typical agentic application reads from full-text, vector, columnar, and session-memory stores in a single turn — over four wire protocols, four vendors, and four bills.

DATA GRAVITY

CONTEXT IS DATA-LOCAL.

Logs, embeddings, metrics, and sessions describe the same events. Joining them in a Python orchestrator is an architecture mistake; the join belongs in storage.

SOVEREIGNTY

YOUR DATA. YOUR PERIMETER.

Enterprise data — security telemetry, customer records, regulated content — does not leave the customer's control. The retrieval engine has to be self-hostable, top to bottom.

DETERMINISM

SLOS THAT HOLD.

Agent loops amplify tail latency by the number of retrieval calls per turn. A 99th-percentile spike upstream is a user-visible stall downstream. Latency must be predictable, not average-good.

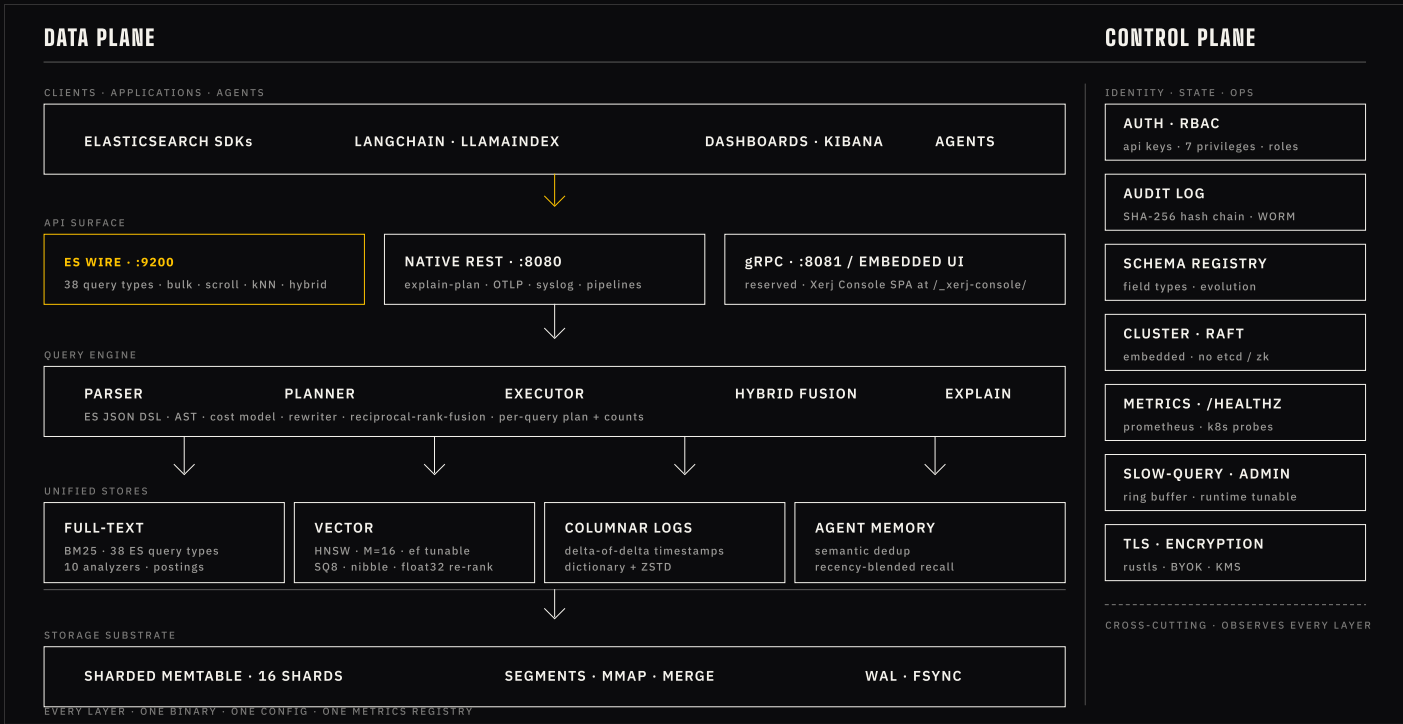
WHAT XERJ ARGUES

The right architectural answer is a unified data plane — one engine that natively understands every retrieval modality, speaks the wire protocol enterprises already use, and runs inside the customer's security boundary.

03 · SYSTEM AT A GLANCE

CONTROL PLANE. DATA PLANE.

Xerj is one binary that exposes one cluster as one engine. The control plane carries cluster state, schema, identity, and audit. The data plane carries ingest, storage, search, and retrieval. Both are observable through one Prometheus endpoint. Both speak the same wire protocols every existing client already speaks.



— DATA PATH — PRIMARY API · :9200 --- CONTROL PATH

KEY PROPERTY · DATA PLANE AND CONTROL PLANE SCALE INDEPENDENTLY

04 · ARCHITECTURAL PRINCIPLES

SIX RULES. ONE ENGINE.

The shape of the engine is a direct consequence of six commitments. Every internal trade-off — sharded memtables, mmap-resident segments, embedded consensus, in-process embedding proxy — is the cheapest implementation of one of these principles.

01

UNIFIED RETRIEVAL

Full-text, dense-vector, columnar logs, and agent memory share a single segment cache, a single write-ahead log, and a single permission model. There is no cross-store orchestrator because there is no cross-store boundary.

02

DROP-IN COMPATIBILITY

The primary API is the Elasticsearch 8.13 wire protocol. Existing clients, dashboards, and ingest collectors connect unchanged. Migration is a configuration change, not a rewrite.

03

DETERMINISTIC RUNTIME

Rust, no garbage collector, no JVM warm-up. Cold start is single-digit milliseconds. Tail latency is governed by code paths, not heap pressure. SLOs measured at p99.9 mean what they say.

04

SOVEREIGN BY DEFAULT

A single static binary. Air-gap installable. No mandatory cloud dependency. Identity, encryption keys, and audit data live wherever the customer's policy says they live.

05

EXPLAINABLE BY CONSTRUCTION

Every query produces an explain-plan with cost and row counts. Every privileged action produces an audit entry. Every release ships with reproduction scripts for every published number.

06

ONE OPERATIONAL SURFACE

One config file. One metrics endpoint. One audit trail. One backup story. The platform that runs SIEM also runs RAG, observability, and agent memory — same on-call rotation, same dashboards, same upgrade procedure.

05 · SYSTEM LAYERS

FROM WIRE TO SEGMENT.

The engine is a stack of well-bounded layers. Wire protocols live at the top. The query planner sits in the middle. The storage substrate lives at the bottom. Each layer has one responsibility and one downstream contract — which is what makes the engine fast to read for a new architect and safe to modify for a contributor.



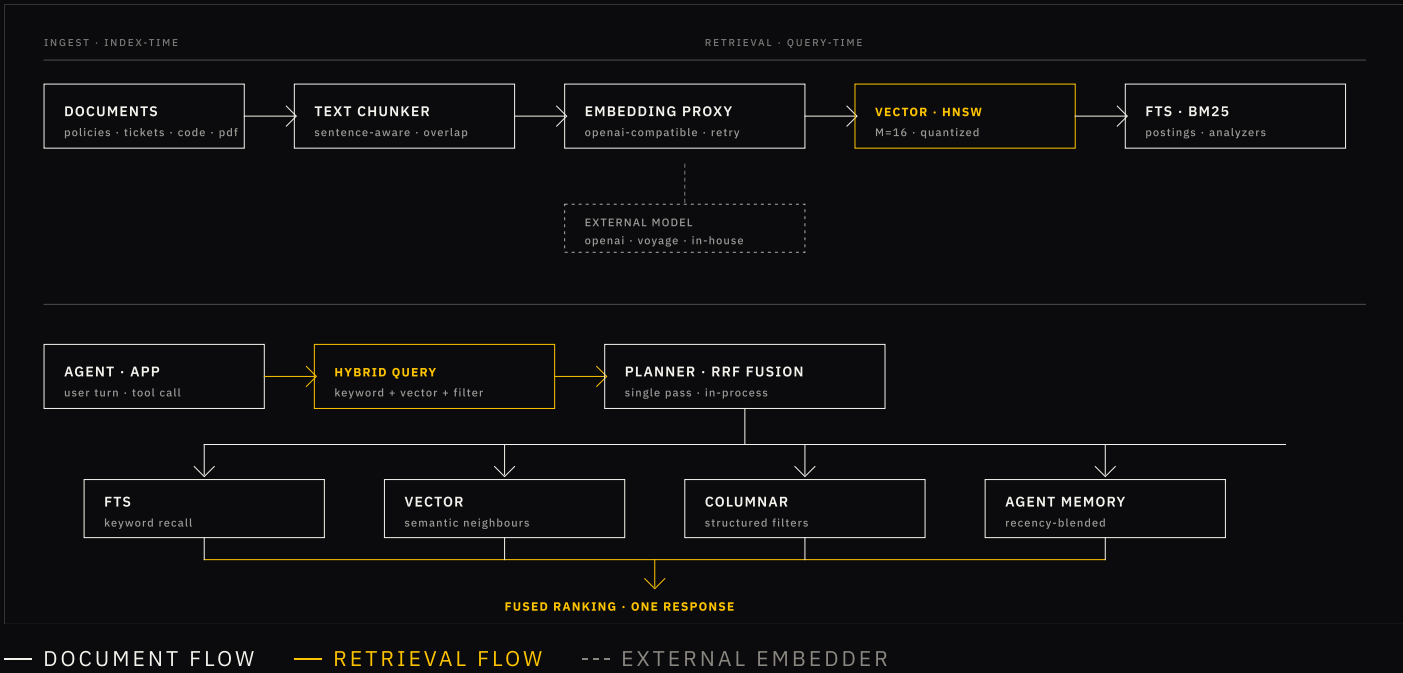
— REQUEST PATH — CRITICAL HOT PATH --- DURABILITY · STATE

EVERY LAYER · OWNS ONE CONCERN · EXPOSES ONE CONTRACT

06 · AI & RAG PLANE

CHUNK. EMBED. RECALL. REASON.

The AI plane turns documents and conversations into retrievable context. Three primitives — chunking, embedding, agent memory — sit in-process with the retrieval engine, so the round trip between the model and the right context is one network hop and one cache.



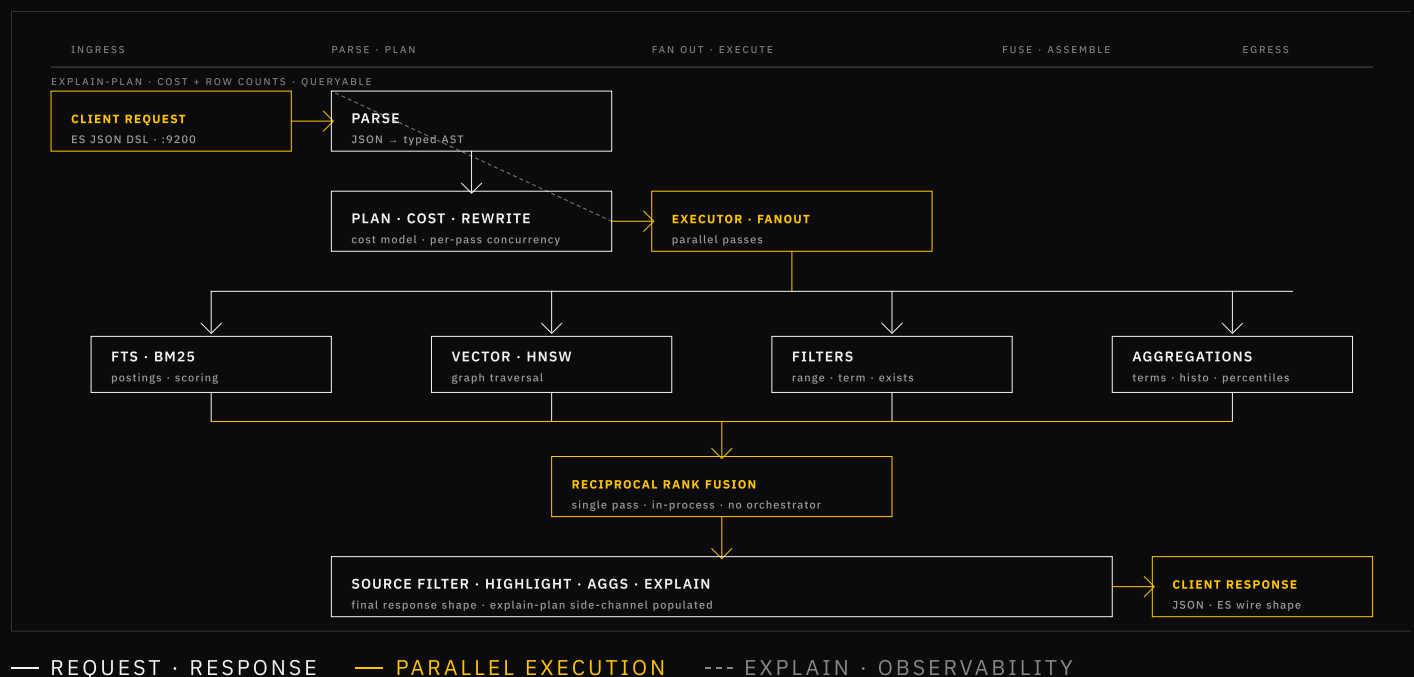
WHAT THIS BUYS THE BUSINESS

CONTEXT QUALITY	Chunking is sentence-aware with configurable overlap, so semantic boundaries survive the split. Recall draws from keyword and vector signals at once, then blends with recency for memory tasks.
VENDOR INDEPENDENCE	The embedding proxy speaks the OpenAI-compatible HTTP API. Use OpenAI today, Voyage tomorrow, an in-house model next quarter — without re-indexing, and without re-architecting the application.
AGENT MEMORY	An append-only memory store with semantic dedup and a recency-blended scoring function. Agents stop re-asking, stop forgetting, and stop drifting across sessions.

07 · HYBRID QUERY LIFECYCLE

PARSE. PLAN. FAN OUT. FUSE.

A hybrid query is the default. One request can express a keyword match, a vector neighbourhood, an aggregation, and a structured filter. The planner decomposes that into independent passes, runs them concurrently against the right index, and fuses the results with reciprocal rank fusion in a single pass — no external orchestrator, no cross-store HTTP calls.

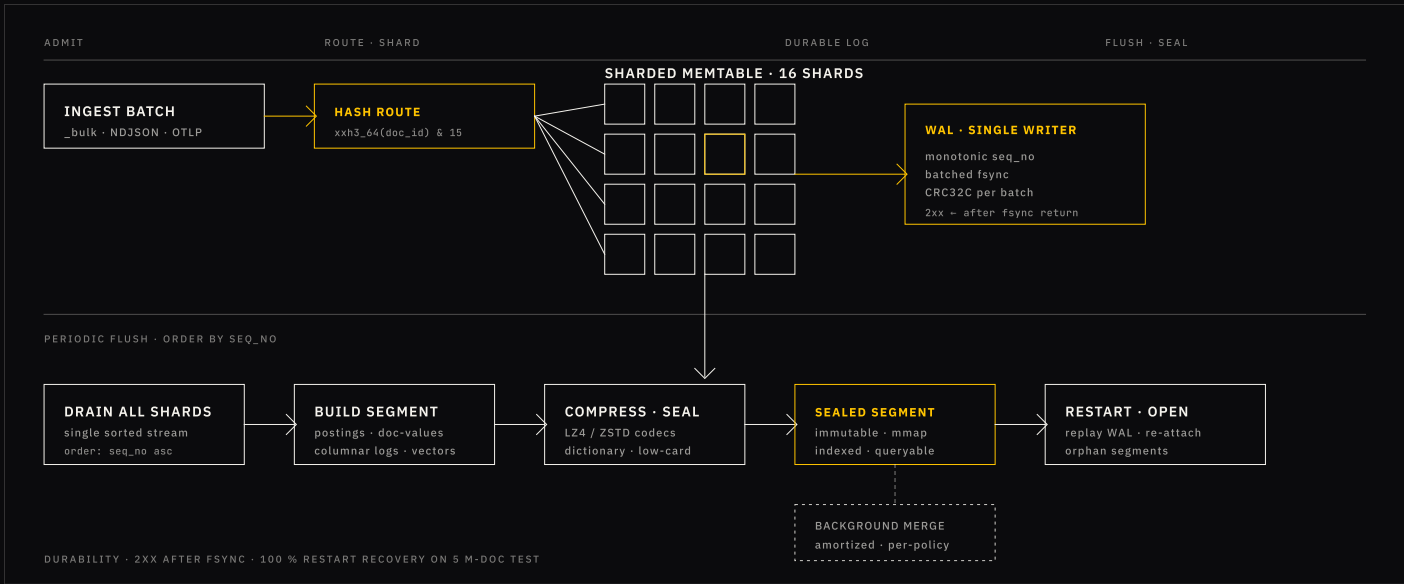


EVERY QUERY · RETURNS AN EXPLAIN-PLAN WITH COST AND ROW COUNTS

08 · STORAGE & DURABILITY

SHARDED MEMTABLE. SEALED SEGMENTS.

Writes hit a sharded in-memory table to spread contention across cores. Durability is anchored in a single write-ahead log with monotonic sequence numbers, fsynced once per batch. Periodic flushes drain the memtable into immutable segments — compressed, memory-mapped, and merged in the background. Restart replays the log; orphan segments are re-attached on open.



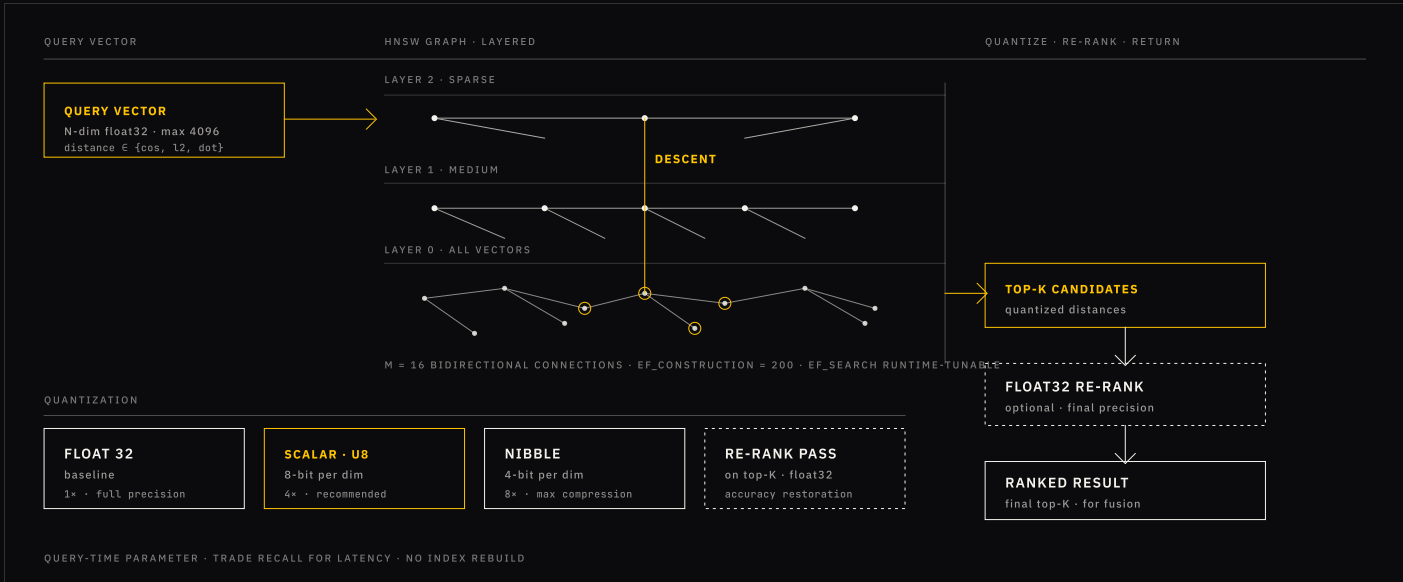
— WRITE PATH — DURABLE STATE --- BACKGROUND TASK

DURABILITY	Every accepted batch is persisted to the WAL with fsync and a per-batch CRC before the client sees a 2xx. Crash recovery replays from the last checkpoint, idempotent per sequence number.
BACK-PRESSURE	When the WAL queue exceeds its watermark, ingest returns 429 with a structured retry hint. Producers slow down; the engine never silently drops.
RECOVERY	Open re-scans the data directory and re-attaches any orphan segments left by a concurrent flush and exit. Restart on a five-million-document workload preserves every accepted document.

09 · VECTOR SUBSYSTEM

HNSW. QUANTIZED. RE-RANKED.

The vector engine is an in-process HNSW graph with three distance metrics, three quantization levels, and an optional full-precision re-ranking pass. Recall and latency are runtime-tunable through a single parameter, so the same index can serve a low-latency conversational lookup and a high-recall offline batch from the same on-disk artifact.



QUANTIZATION

Three modes: float32 baseline, scalar 8-bit (4× compression), nibble (8× compression). The HNSW traversal uses the quantized form; an optional float32 re-ranking pass restores precision on the final top-K.

DISTANCE METRICS

Cosine, L2, and dot product. Per-index choice. The metric is part of the on-disk format header; a loader rejects any index with a mismatched metric or format version.

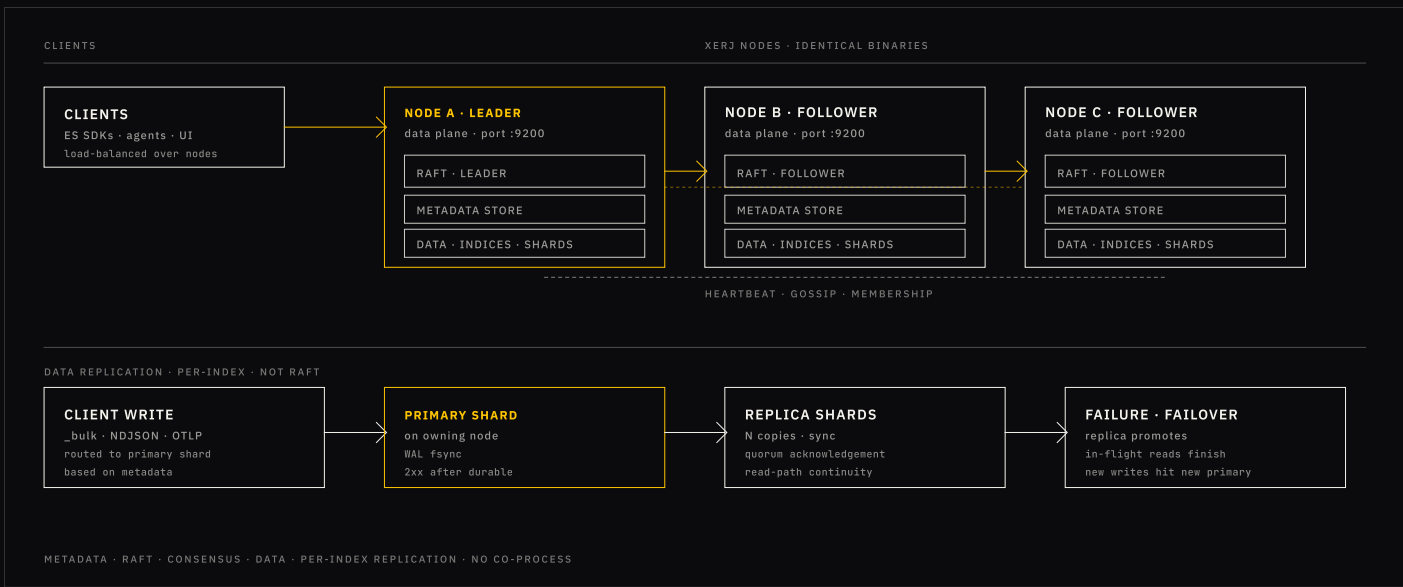
RECALL · LATENCY DIAL

A single runtime parameter trades recall for latency without rebuilding the index. Low-latency UIs and high-recall analytics share one artifact.

10 · CLUSTER & STATE

EMBEDDED CONSENSUS. NO CO-PROCESS.

Cluster metadata — index schemas, shard assignments, the node roster — is replicated by an embedded Raft state machine. There is no separate ZooKeeper, etcd, or Consul to deploy, license, page on, or upgrade in lock-step with the engine. The data plane stays out of the consensus path: only metadata changes take a Raft round-trip.



— DATA TRAFFIC — RAFT (METADATA) --- HEARTBEAT · GOSSIP

METADATA SCOPE	Indices, schemas, shard placement, node membership. Replicated by Raft. Reads of metadata never block on consensus; the leader acknowledges, and followers serve.
DATA SCOPE	Documents, segments, vectors, and logs flow over the data ports. They are not Raft-replicated — replication for durability is per-index, governed by the metadata that Raft owns.
FAILURE MODEL	A leader loss costs one election round-trip on the metadata plane only. Reads and writes against existing indices continue uninterrupted.

11 · SECURITY & COMPLIANCE

TAMPER-EVIDENT. RBAC-NATIVE.

Every privileged operation is a journal entry in an append-only audit log whose entries chain by SHA-256. Tampering with any entry breaks the chain at that point and at every subsequent entry. A single endpoint verifies the chain end-to-end; operators pin known-good chain heads externally for cross-restart attestation.



— JOURNAL ENTRY — SHA-256 LINK --- VERIFY · EXPORT

CONTROL	IMPLEMENTATION	STATUS
API-KEY AUTH	Admin key generated on first start; subsequent keys minted under RBAC. Auth on every endpoint.	GA
RBAC	Seven privileges; six seeded roles (admin, write, read, read-only-index, snapshot-admin, auditor). Per-pattern index scoping.	BETA
AUDIT LOG	Append-only, hash-chained, queryable. Verify endpoint walks the chain end-to-end.	GA
TLS · TRANSIT	rustls-based listener, modern ciphers. In-process integration scaffolded; final wiring lands at v1.0.0.	BETA
ENCRYPTION AT REST	AES-GCM on segment + WAL files; keys held by an external KMS or BYOK-style integration.	ROAD V1.0

12 · OPERATIONS & OBSERVABILITY

ONE BINARY. DAY-TWO READY.

Day-two operations are baked in. Liveness and readiness probes ship on both API ports for Kubernetes. A built-in slow-query log surfaces regressions before they reach production. The embedded admin UI gives an operator everything an SRE needs without installing a second product.

OPERATIONAL SURFACE

CAPABILITY	SHAPE	STATUS
LIVENESS · READINESS	Standard Kubernetes probes on both data-plane ports. Listeners only bind after WAL replay; readiness then tracks cluster health — rolling restart is fleet-safe by default.	GA
METRICS	Prometheus exposition. Counters, histograms, and gauges across ingest, query, flush, cache. Per-instance registry — no global-singleton label collisions.	GA
SLOW-QUERY LOG	Bounded ring buffer of slow queries, threshold runtime-tunable. List, clear, and re-threshold from a single admin endpoint.	GA
EMBEDDED UI	The Xerj Console playground SPA is compiled into the binary and served at a reserved path. Zero external dependency to operate the cluster.	GA
EXPLAIN PLAN	Per-query cost and row counts queryable as a first-class endpoint. Quotas and circuit breakers enforce per-tenant ceilings before execution.	GA
HELM & DOCKER	StatefulSet chart with PVC, headless service, non-root security context, multi-arch image, HEALTHCHECK wired to readiness.	GA
SLOW · ADMIN APIS	List, clear, threshold-set on the slow-query buffer; audit search and verify; security role list and put.	GA

13 · MEASURED PERFORMANCE

REAL HARDWARE. REAL WORKLOAD.

The numbers below come from the v1.0.0-rc.1 verification run on a 32-core NVMe host against a real OpenSSH log corpus and a thousand-vector RAG workload. Every comparison number against Elasticsearch 8.13 was measured on the same host with identical wire payloads — not extrapolated.

COLD START 4 ms VS ES 6.05 S · 863×	BURST INGEST 5.97 M DOCS/SEC · LOGHUB OPENSSSH	SUSTAINED 1.43 M DOCS/SEC · 5 M-DOC RUN	BINARY 19.6 MB SINGLE STATIC · NO JVM
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HEAD-TO-HEAD VS ELASTICSEARCH 8.13

DIMENSION	ES 8.13	XERJ	DELTA	SOURCE
Cold start	6.05 s	4 ms	863×	verify run
RSS · 1 K-vector index	2,857 MB	727 MB	3.9×	verify run
Index creation	93.9 ms	2.4 ms	39×	verify run
Term query · p50	0.79 ms	0.31 ms	2.5×	verify run
Match query · p50	1.28 ms	0.33 ms	3.9×	verify run
Top-K sort · p50	4.28 ms	0.34 ms	12.6×	verify run
Bulk · 1 K vectors	10,219 d/s	22,974 d/s	2.25×	verify run

WIRE COMPATIBILITY

ES YAML CONFORMANCE	1,305 / 1,329 cases pass — 98.2 % against the published Elasticsearch 8.13 REST conformance suite. Existing clients, dashboards, and ingest collectors connect unchanged.
DURABILITY UNDER RESTART	Five-million-document workload: every accepted document survives a process restart. Orphan-segment recovery on open closes the historical concurrent-flush-and-exit data-loss window.

14 · GET STARTED

PROVE IT ON YOUR DATA.

The fastest path is a two-week pilot on a real workload. We send the binary, the Helm chart, and the reproduction scripts for every number in this brief. You send one workload — SIEM, RAG, observability, or agent memory. We meet again in two weeks with your numbers, on your hardware, behind your perimeter.

CONTACT

HELLO@XERJ.AI

XERJ.AI · XERJ.COM · XERJ.DEV

PILOT PROCESS

WEEK 0	Scoping call · pick workload · sign mutual NDA if needed (not required to receive the binary).
WEEK 1	Deploy in your environment · reproduce the published numbers on sample data · smoke-test ES client compatibility.
WEEK 2	Shadow against your workload · compare latency, footprint, cost. Decision meeting: your numbers, your hardware, your call.